SKILLBRIDGE – SKILL DEVELOPMENT & COMPETENCY MANAGEMENT

A PROJECT REPORT

Submitted by

PREETHI KN

(240701398)

in partial fulfilment of the End Semester Examination for the course

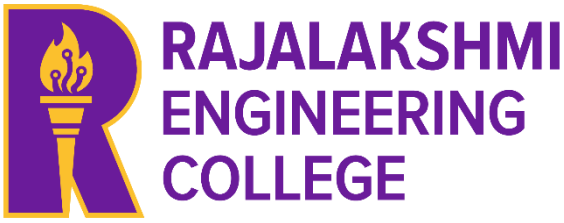
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "SKILLBRIDGE–SKILL DEVELOPMENT & COMPETENCY MANAGEMENT" is the bonafide work of PREETHI KN (240701398), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester V, under my supervision.

SIGNATURE

Dr.J.MANORANJANI
HEAD OF THE DEPARTMENT
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

Mr.BHUVANESWARAN B
COURSE FACULTY
Assistant Professor (SG)
Department of Computer Science and Engineering
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, PREETHI KN (240701398), hereby declare that the project report titled "SKILLBRIDGE–SKILL DEVELOPMENT & COMPETENCY MANAGEMENT", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 30 October 2026

SIGNATURE OF THE STUDENT
PREETHI KN
240701398

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman MR.S.MEGANATHAN , Chairperson DR.M.THANGAM MEGANATHAN and Vice-Chairman MR.ABHAY MEGANATHAN for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr. SN.MURUGESAN, Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. J. MANORANJANI, Head of the Department, Department of Computer Science and Engineering, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, MR. BHUVANESWARAN B, Assistant Professor (SG), for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation / Zoho Creator Education Programme for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

PREETHI KN
240701398

ABSTRACT

The process of planning courses, managing learning resources, coordinating tutors and students, and tracking learning activities can become complex when information is maintained manually or across separate systems. Such fragmented processes may lead to data duplication, delayed communication, difficulty in accessing course information, and increased administrative effort. Students may also face challenges in identifying suitable learning resources and obtaining timely assistance with their academic queries. To address these challenges, SkillBridge was developed as a centralized learning and course management platform using a low-code development approach.

The application was developed using Zoho Creator, which provides form creation, report generation, workflow automation, and application management capabilities. The system includes modules for managing student and tutor information, courses, learning materials, subscriptions, and tutoring requests. Custom forms and reports support data entry, retrieval, and monitoring, while Deluge scripting and workflows automate selected operations and reduce repetitive manual activities. Zoho CRM integration was implemented to support the AI assistant functionality, enabling users to submit learning-related questions and receive AI-generated responses through the configured integration. The application is designed to support different user roles and organize access to relevant features according to their responsibilities

The completed application was evaluated through functional testing of its forms, reports, workflows, and AI integration to verify that the implemented features operated as expected. The system provides a centralized environment for managing learning-related information and improves the accessibility of course resources and assistance. By automating selected tasks and integrating AI-based question answering, SkillBridge has the potential to reduce administrative effort, simplify information retrieval, and improve the responsiveness of learning support. Quantitative improvements in processing time and manual effort require further measurement under actual usage conditions. The project demonstrates how low-code development and CRM integration can be combined to create an adaptable learning management solution.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Zoho CRM Integration; AI-Assisted Learning*

PROJECT AT A GLANCE

Attribute	Details
Project Title	SkillBridge Skill development & Compentency Management
Application Domain	Education
Student Name & Register No.	Preethi KN – 240701398
Zoho Creator Edition	Free
Application Name (Link Name)	SkillBridge
Published / Live URL	https://skillbridge.zoho.in/
Demo Credentials (per role)	Administrator, Tutor and Student
Number of Forms / Subforms	10 Forms / No Subforms
Number of Reports (by type)	28 Reports
Number of Pages / Dashboards	2 Pages
Number of Workflows	17 Workflows
Deluge Functions / Custom Actions	1standalone script for AI integration
Integrations	Zoho CRM, OpenAI Chat
Roles & Profiles Defined	Student, Tutor and Administrator-oriented functionality appears in reports and workflows; verify actual configured permissions and profiles
Environments Used	Zoho Creator Environment
Date of Production Deployment	30/10/2026

TABLE OF CONTENTS

Formatting Standards (Anna University Convention) .....**Error! Bookmark not defined.**

Working with the Template ..... **Error! Bookmark not defined.**

ABSTRACT.....iv

PROJECT AT A GLANCE v

LIST OF TABLES.....ix

LIST OF FIGURES x

LIST OF ABBREVIATIONS.....xi

INTRODUCTION 1

1.1 Background.....1

1.2 Problem Statement.....1

1.3 Objectives 1

1.4 Scope of the Project 1

1.5 Justification for a Low-Code Approach.....2

1.6 Organisation of the Report.....2

LITERATURE SURVEY AND SYSTEM STUDY 3

2.1 Study of the Existing System.....3

2.2 Limitations of the Existing System.....3

2.3 Low-Code versus Traditional Development.....3

2.4 Literature Survey 4

2.5 Emerging Trends in Low-Code Development.....5

2.6 Proposed System.....5

REQUIREMENT ANALYSIS 6

3.1 Stakeholders and User Roles 6

3.2 Functional Requirements 6

3.3 Non-Functional Requirements.....7

3.4 Hardware and Software Requirements 7

3.5 Use Case Model 7

SYSTEM DESIGN 9

4.1 System Architecture.....9

4.2 Data Model.....9

4.3 Form Design.....11

4.4 Process and Workflow Design.....11

4.5 User Interface and Navigation Design.....11

4.6 Access Control Design.....12

IMPLEMENTATION 13
5.1 Application Setup 13
5.2 Forms and Data Capture 13
5.3 Reports 14
5.4 Pages and Dashboards 15
5.5 Deluge Scripting 15
5.6 Workflows and Automation 16
5.7 Integrations 17
5.8 User Management and Role-Based Access 18
5.9 AI and Analytics Features 18
5.10 Pro-Code Extension 19
5.11 Mobile and Portal Access 19
TESTING AND DEBUGGING 20
6.1 Testing Strategy 20
6.2 Test Cases 20
6.3 Debugging Deluge Scripts 21
6.4 Defect Log 22
6.5 User Acceptance Testing 22
DEPLOYMENT, SECURITY AND GOVERNANCE 23
7.1 Environments and Deployment Strategy 23
7.2 Performance Optimisation 23
7.3 Security Features 24
7.4 Governance and Administration 24
RESULTS AND DISCUSSION 26
8.1 Application Output 26
8.2 Achievement of Objectives 27
8.3 Comparison with the Existing System 28
8.4 Limitations of the Low-Code Approach 28
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum 29
CONCLUSION AND FUTURE ENHANCEMENTS 30
9.1 Conclusion 30
9.2 Future Enhancements 30
9.3 Learning Outcomes and Career Relevance 31
REFERENCES 32
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST 33
APPENDIX B – COURSE OUTCOME MAPPING 36
APPENDIX C – COMPLETE DELUGE SOURCE CODE 37

APPENDIX D – USER MANUAL 38
D.1 Accessing the Application 38
D.2 Instructions for [Role 1] 38
D.3 Instructions for [Role 2] 38
APPENDIX E – EVALUATION RUBRIC 39
APPENDIX F – VIVA-VOCE RECORD SHEET 41

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application....3
Table 2.2: Summary of Literature Survey....4
Table 3.1: Stakeholders and User Roles....6
Table 3.2: Functional Requirements....6
Table 3.3: Non-Functional Requirements....7
Table 3.4: Hardware and Software Requirements....7
Table 3.5: Use Case Description....8
Table 4.1: Entities and Relationships....10
Table 4.2: Field Specification – Form....11
Table 4.3: Role–Permission Matrix....12
Table 5.1: Reports Implemented....14
Table 5.2: Deluge Scripts Implemented....15
Table 5.3: Workflows Implemented....16
Table 5.4: Integrations Implemented....17
Table 6.1: Test Cases....20
Table 6.2: Defect Log....22
Table 6.3: User Acceptance Feedback....22
Table 7.1: Deployment Checklist....23
Table 7.2: Performance Optimisation Results....24
Table 7.3: Security Controls Implemented....24
Table 8.1: Mapping of Objectives to Outcomes....27
Table 8.2: Existing System versus Proposed Application....28
Table A.1: Feature Coverage Checklist....33
Table B.1: Mapping of Course Outcomes to Report Chapters....36
Table E.1: Evaluation Rubric for the Individual Project....39
Table E.2: Performance Level Descriptors....40
Table F.1: Viva-Voce Questions and Responses....41
Table F.2: Marks Summary....41

LIST OF FIGURES

Figure 2.1: Process Flow of the Existing System 3
Figure 3.1: Use Case Diagram of the Application 8
Figure 4.1: System Architecture of the Application ... **Error! Bookmark not defined.**
Figure 4.2: Entity–Relationship Diagram 10
Figure 4.3: Workflow / Process Flow for Tutoring Request 11
Figure 4.4: Navigation Structure and Wireframe of Dashboard 12
Figure 5.1: Application Home and Navigation Sections 13
Figure 5.2: Add course Form in the Form Builder 14
Figure 5.3: Add course Form in Live Mode 14
Figure 5.4: All Course Material Report 14
Figure 5.5: Dashboard Page Name 15
Figure 5.6: Workflow Configuration 16
Figure 5.7: Connection Setup and Postman / Response Output 17
Figure 5.8: Roles and Permission Profiles Configuration 18
Figure 5.9: AI / Analytics Feature in Use 19
Figure 5.10: Application on Mobile / Customer Portal 19
Figure 6.1: Deluge Debug Output / Log 22
Figure 7.1: Environments / Publish to Production 23
Figure 7.2: Audit Trails 25
Figure 8.1: Scenario Step 1 – Course Management 26
Figure 8.2: Scenario Step 2 – Course Enrolment 27
Figure 8.3: Scenario Step 3 – AI Assistant 27

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

SkillBridge: Skill Development and Competency Management is a low-code application developed using Zoho Creator. It aims to simplify course management, learning-material organisation, student enrolment, subscription tracking, and tutoring support. The application includes an AI learning assistant through the project's Zoho CRM integration. It uses forms, reports, pages, workflows, and Deluge scripting to manage learning-related activities in one place.

1.2 Problem Statement

Managing courses, learning materials, enrolments, and tutoring requests separately can lead to difficulties in organising information and tracking activities. SkillBridge addresses these challenges by providing a centralised application.

- Difficulty organising course and learning-material information.
- Challenges in tracking and enrolments and subscriptions.
- Lack of structured tutoring-request management.
- Need for centralised reports and AI-assisted learning support.

1.3 Objectives

- To develop forms for managing courses, topics, materials, and enrolments
- To automate selected processes using Deluge scripting and workflows.
- To provide reports and pages for viewing application information.
- To integrate AI-assisted learning functionality through Zoho CRM.
- To test the application and identify areas for improvement.

1.4 Scope of the Project.

The project includes course management, learning materials, enrolment, subscription and payment records, tutoring requests, reports, and AI learning assistance. It is developed using Zoho Creator.

Advanced examinations, verified live payment processing, a standalone mobile application, and enterprise-level security features are outside the confirmed scope unless separately implemented.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected because it supports rapid application development through visual forms, reports, workflows, and Deluge scripting. It reduces the need to develop every component from scratch and provides integration and cloud-hosting capabilities. These features make it suitable for developing and maintaining SkillBridge efficiently

1.6 Organisation of the Report

Chapter 1 introduces the project. Chapter 2 reviews existing systems and related literature. Chapter 3 presents the requirements. Chapter 4 explains the system design. Chapter 5 describes implementation. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security, and governance. Chapter 8 presents the results and discussion. Chapter 9 concludes the report and suggests future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a conventional learning-management process, course details, learning materials, student enrolments, subscriptions, and tutoring requests may be handled using separate tools or manual records. This can make information management and tracking more difficult. SkillBridge addresses these needs through a centralised Zoho Creator application.

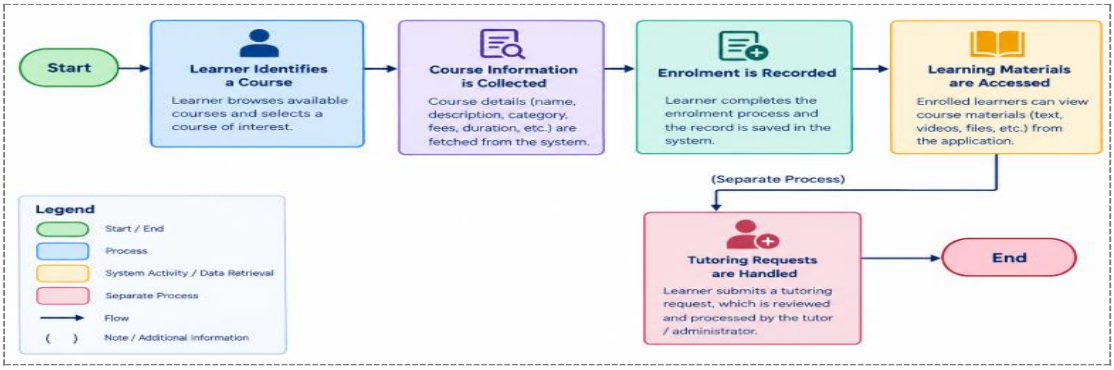


Figure 2.1: Process Flow of the Existing System

2.2 Limitations of the Existing System

- Enrolment and subscription details can be difficult to track.
- Course and learning-material information may be scattered across different records.
- Tutoring requests may lack a centralised tracking process.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	8–10 weeks	2–3 weeks
Skills required	Programming, database and deployment skills	Platform knowledge and basic scripting
Hosting & infrastructure	Infrastructure must be arranged	Platform-managed hosting options

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Mobile application	Usually requires additional development	Mobile access options are available
Security & access control	Implemented and configured by the development team	Platform features plus application configuration
Maintenance & change	Code changes may be extensive	Forms and workflows can be modified visually
Customisation limits	Greater control over implementation	Depends on platform capabilities and extensions
Cost	Depends on development and infrastructure	Depends on subscription, usage and development effort

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Bucaioni et al., 2022	*Modelling in Low-Code Development: A Multi-Vocal Systematic Review*	Reviews low-code approaches and identifies benefits such as abstraction and automation.	Supports the choice of low-code development.
2	Frank, Maier and Bock, 2021	*Low Code Platforms: Promises, Concepts and Prospects*	Compares low-code platforms and their capabilities.	Helps evaluate the development approach.
3	Zoho Corporation	Zoho Creator Features	Describes forms, reports, workflows, integrations and hosting options.	Supports the selection of Zoho Creator.
4	Moodle	Moodle Learning Platform	Provides learning-management features for organising learning activities.	Offers a comparison with learning platforms.
5	Coursera	Online Learning Platform	Provides online courses and subscription-based learning options.	Provides a reference for course and subscription functionality.

2.5 Emerging Trends in Low-Code Development

Important trends include AI-assisted application building, citizen development, workflow automation and integration of modular services. Zoho Creator provides visual application-building tools and integration capabilities [1]. These trends influenced SkillBridge's use of forms, workflows, reports and an AI learning-assistant integration through Zoho CRM. They also provide opportunities for future improvements in automation and analytics.

2.6 Proposed System

SkillBridge is a centralised skill-development application built using Zoho Creator. It provides modules for course management, learning materials, enrolments, subscriptions, payment-related records, tutoring requests and AI-assisted learning.

The proposed system addresses fragmented information by organising records within one application, supports structured tutoring-request management, and provides reports and pages for viewing information. Deluge scripting and workflows support selected application processes. Further testing and appropriate security configuration are necessary before making claims about production readiness.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Manages the application	Manage records and oversee application settings
Tutor	Supports learning and tutoring	Handle tutoring requests and related approvals
Student	Uses learning facilities	Access course information, materials and tutoring support
Application Operator	Maintains application records	Create and update relevant records

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow users to manage course details, categories and topics.	Administrator	High
FR-02	The system shall allow learning materials to be recorded and viewed.	Administrator, Student	High
FR-03	The system shall maintain student enrolment records.	Administrator, Student	High
FR-04	The system shall store subscription and payment-related records.	Administrator, Student	Medium
FR-05	The system shall record tutoring requests and related approval details.	Administrator, Student, Tutor	High
FR-06	The system shall display relevant information through reports and pages.	Administrator, Users	Medium
FR-07	The system shall provide AI-assisted learning functionality through the configured integration.	Student	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Forms and reports should respond within a reasonable time under normal usage.
NFR-02	Security	Access to records should follow the configured user permissions.
NFR-03	Usability	The interface should be clear and easy to navigate.
NFR-04	Availability	The application should be accessible when the platform and required services are available.
NFR-05	Maintainability	Forms and workflows should be easy to update and maintain.
NFR-06	Auditability	Relevant record changes should be traceable where audit features are configured.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Integration	Configured AI service
Other Zoho services	Zoho CRM
Mobile	Zoho Creator mobile app (Android)
Client hardware	Intel Core i3/i5 or above, 16 GB RAM, internet connection

3.5 Use Case Model

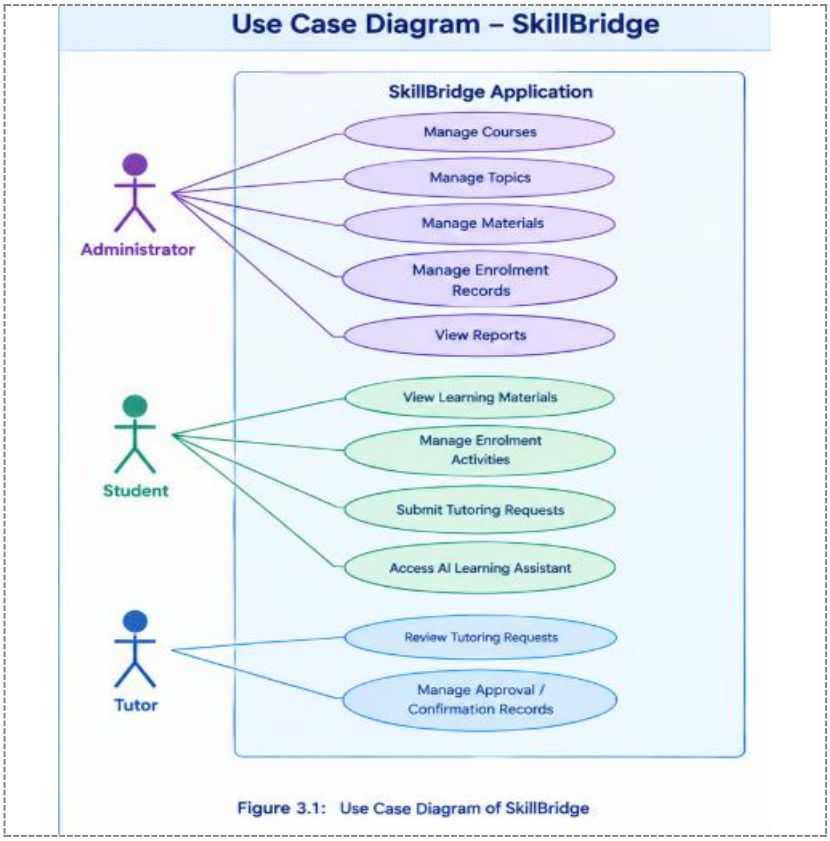


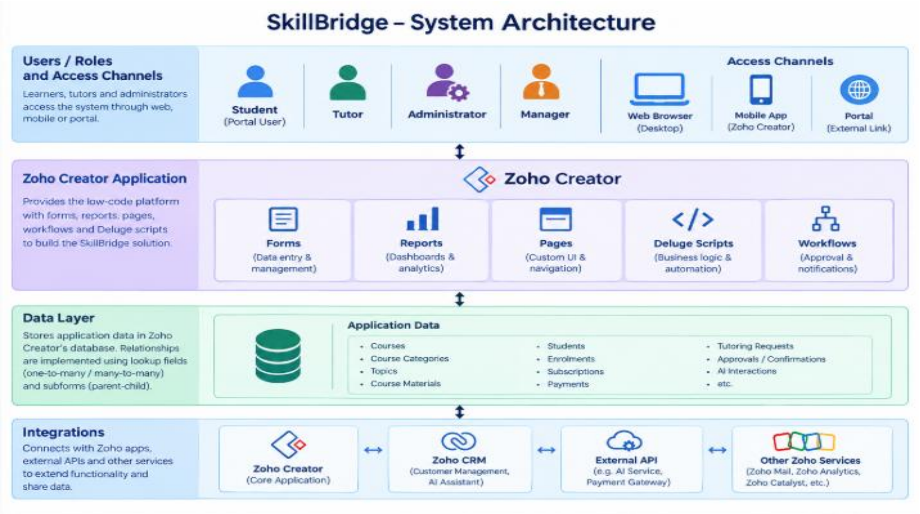
Figure 3.1: [Use Case Diagram of the Application]

Table 3.5: Use Case Description

Use Case ID	UC-01
Use Case Name	Submit Tutoring Request
Actor(s)	Student
Pre-condition	The student can access the relevant application form.
Main Flow	1. Student opens the tutoring-request form. 2. Student enters the required details. 3. Student submits the form. 4. The request is recorded if validation succeeds.
Alternate Flow	If required information is missing or invalid, the form should request correction where validation is configured.
Post-condition	The tutoring request is stored for further handling.
Related FR	FR-05

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture



The diagram illustrates four main layers:

1. Users and Access: Students, tutors, administrators, and managers access the application.
2. Application Layer: Zoho Creator provides forms, reports, pages, Deluge scripts, and workflows.
3. Data Layer: Stores course categories, courses, topics, materials, enrolments, subscriptions, payment records, tutoring requests, and related information.
4. Integration Layer: Connects Zoho Creator with Zoho CRM and configured external services, including AI functionality.

4.2 Data Model

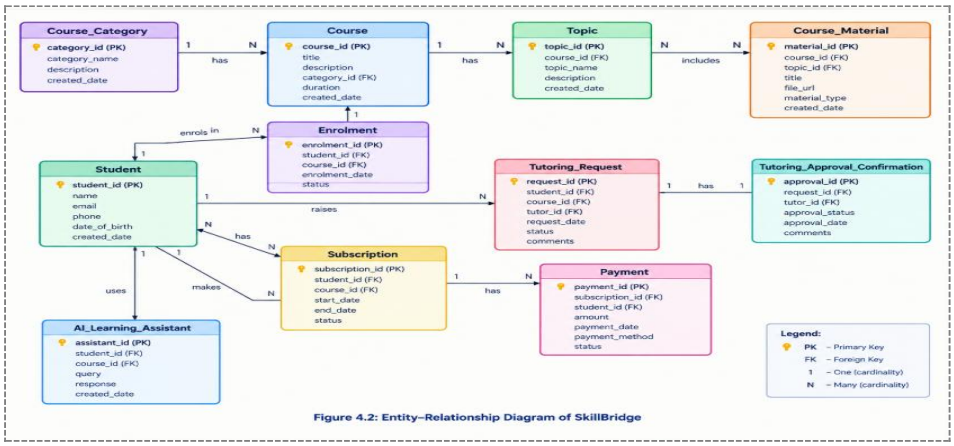


Figure 4.2: Entity–Relationship Diagram of SkillBridge

Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Add_Course_Category	Category ID, Category Name, Description	Add_Course	One-to-Many	Lookup in Add_Course
Add_Course	Course ID, Course Name, Description, Category	Add_Course_Category, Add_Topic, Enroll_User	One-to-Many	Lookup fields
Add_Topic	Topic ID, Topic Name, Course	Add_Course, Add_Course_Material	One-to-Many	Lookup fields
Add_Course_Material	Topic ID, Topic Name, Course	Add_Course, Add_Course_Material	One-to-Many	Lookup fields
Enroll_User	Enrolment ID, Student, Course, Status	Add_Course, Student/User	Many-to-One	Lookup fields
Tutoring_Request	Request ID, Student, Course, Status	Student/User, Add_Course, Tutoring_Approval_Confirmation	One-to-One or One-to-Many	Lookup fields
AI_Learning_Assistant	Record ID, Query, Response	Student/User	Many-to-One or One-to-Many	Lookup fields

4.3 Form Design

Table 4.2: Field Specification – [Form Name] Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Course ID	Course_ID	Auto Number	No	System generated	Unique course identifier
Course Name	Course_Name	Single Line	Yes	Must not be empty	Name of the course
Category	Course_Category	Lookup	Yes	Select an existing category	Links to Add_Course_Category
Duration	Duration	Number	No	Must be a positive number	Course duration
Created Date	Created_Date	Date-Time	No	Current date and time, if configured	Record creation information

4.4 Process and Workflow Design

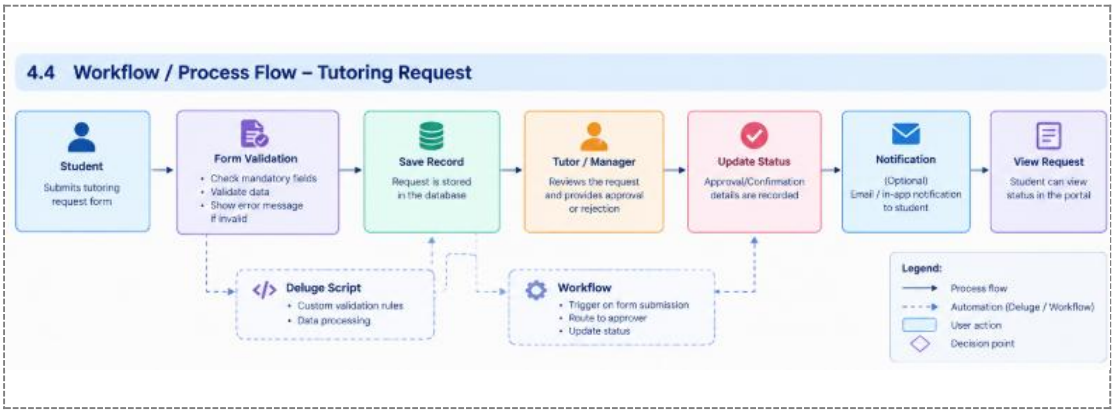


Figure 4.3: Workflow / Process Flow for Tutoring Request

4.5 User Interface and Navigation Design

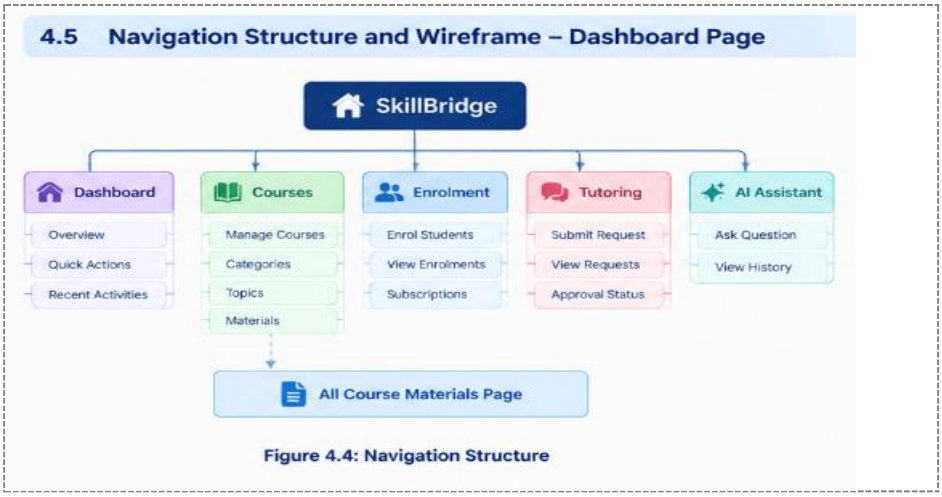


Figure 4.4: Navigation Structure and Wireframe of Dashboard Page

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Administrator	Manager	Staff	Portal User
Form:Add_Course	A E V D	A E V	V	V
Form: Add_Course_Category	A E V D	A E V	V	V
Report: Course / Enrolment Reports	V	V	V	V
Page:Dashboard	V	V	V	V

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

SkillBridge was developed using Zoho Creator, a low-code application development platform. The application provides a centralised environment for managing courses, learning materials, enrolments, subscriptions, tutoring requests and AI-assisted learning.

The application contains forms for data entry, reports for viewing records, and pages for presenting information. Deluge scripting and workflows support selected application processes. The application includes the Dashboard and All_Course_Materials_Page pages.

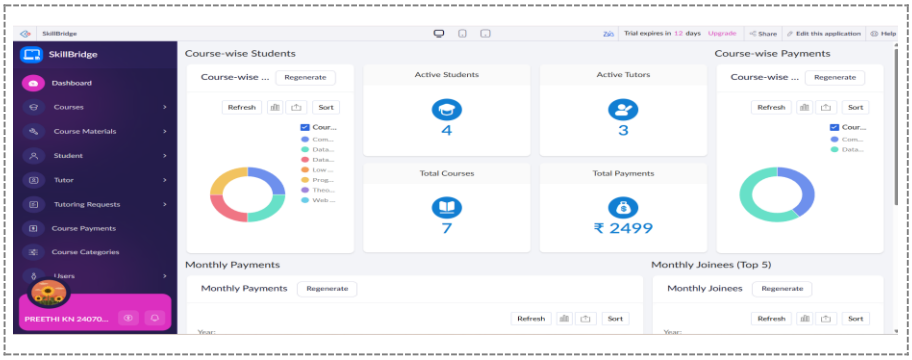


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Forms are the primary components used to collect and manage information in SkillBridge. Each form represents a particular application module. Field types, mandatory settings, lookup relationships and validation rules should be configured according to the data requirements.

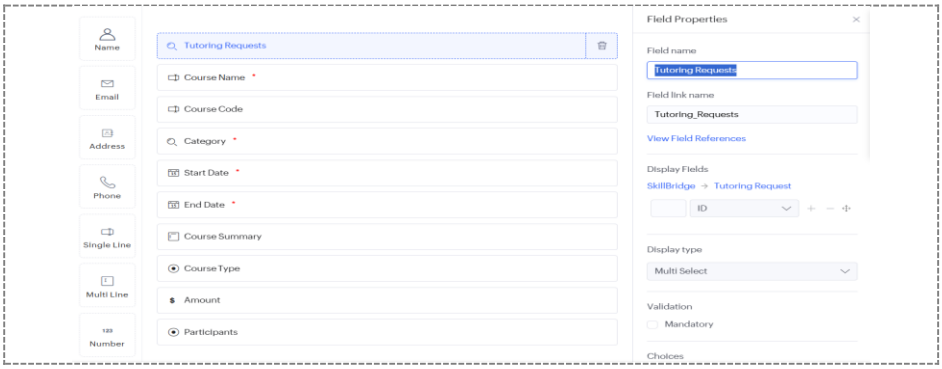


Figure 5.2: Add Course Form in the Form Builder

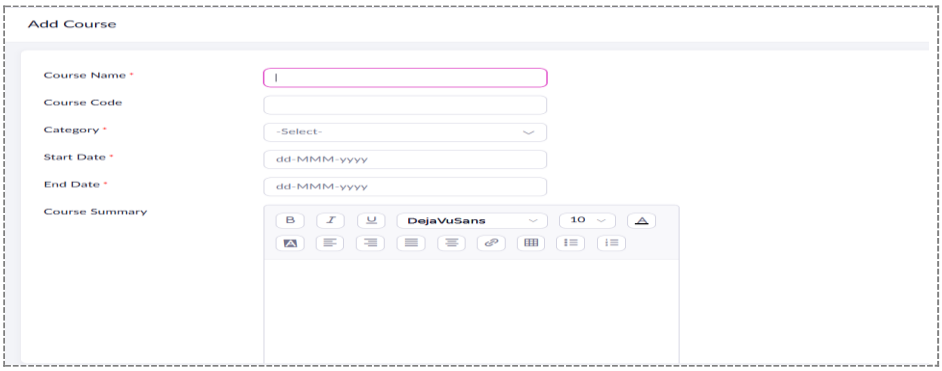


Figure 5.3: Add course Form in Live Mode

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Course Materials	List	Add_Course_Material	Group by Status	Materials displayed	Admin, Student, Tutor
Course Reports	Kanban	Add_Course	Columns by Status	Visualization	Admin, Student, Tutor
Course Category Reports	List	Add_Course_Category	Group by Category	Shows category (Lab/Theory/LOT)	Admin, Student, Tutor
Topic Reports	List	Add_Topic	Filtered by topics	Topics are displayed	Admin, Student, Tutor
Enrolment Reports	List	Enroll_User	Filtered as student/Tutor	Users are displayed	Admin

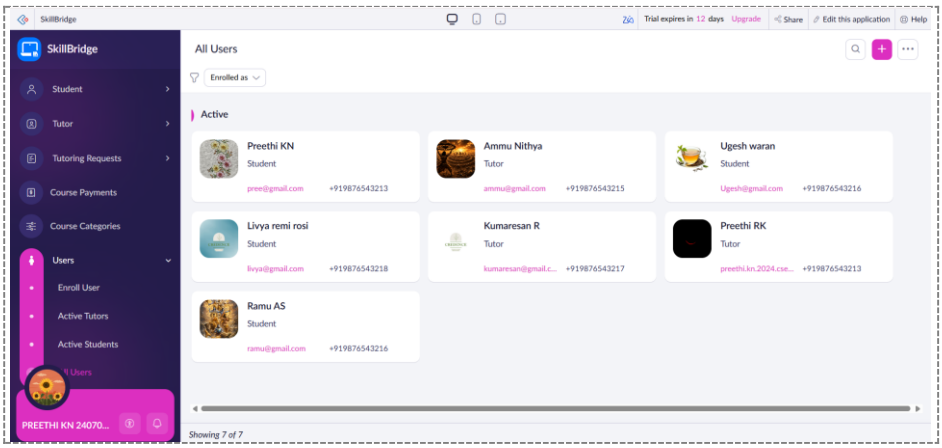


Figure 5.4: All users Report

5.4 Pages and Dashboards

The **Dashboard** provides a page for presenting relevant application information. Its purpose is to offer convenient access to selected records and application features.

The **All_Course_Materials_Page** provides a page for displaying or accessing course-material information.

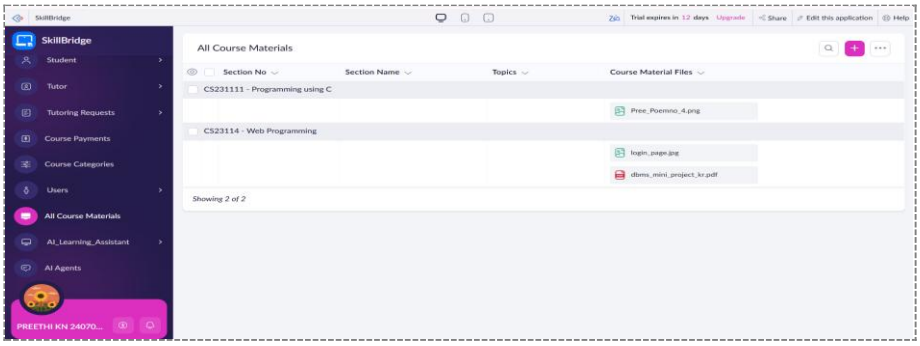


Figure 5.5: All course material Page

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Standalone Function: Add_AI_Agent_Microservice	Inserts an agent-configuration message into the AI Learning Assistant form.	Function definition, record insertion, field assignment
DS-02	Form Workflow: Generate_AI_Response	Handles the configured AI response-related process.	Form workflow, input fields, conditional logic.
DS-03	Other Form Workflows	Supports the configured form-level operations in SkillBridge.	Form events, record handling, Deluge statements
DS-04	Schedule Definition	Executes the scheduled operation configured in the application.	Scheduling the period of course

Code 5.1: Validation script

```
if(input.Course_Name == null || input.Course_Name.trim() == "")
{
   alert "Please enter the course name.";
   cancel submit;
}
```

Code 5.2: Notification script

```
insert into AI_Learning_Assistant
[
   Query = "Provide learning assistance"
   Response = "AI learning assistant configuration initialized."
];
```

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Enrolment Validation	Form Workflow	On Validate	Required enrolment details are missing	Display validation message and block submission	DS-01
Tutoring Request Review	Form Workflow	On record creation	A new tutoring request is submitted	Store the request for review and approval	DS-02
Tutoring Approval	Form Workflow	On approval or record update	Request status is updated	Update the approval/confirmation record	DS-03
AI Learning Assistant	Form Workflow	On form submission	User submits a learning query	Process the AI assistant response workflow	Generate_AI_Response

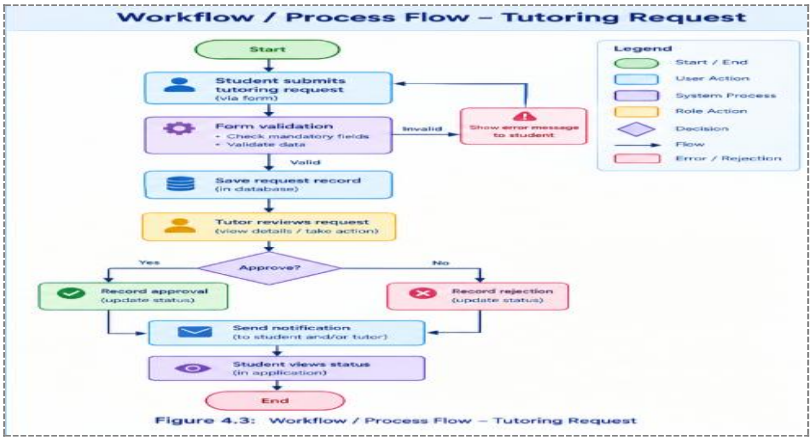


Figure 5.6: Workflow Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho CRM Integration	Zoho CRM – AI assistent	CRM integration / API operation	OAuth 2.0	Supports the reported integration between SkillBridge and Zoho CRM. Exact records exchanged should be verified.
AI Learning Assistant	Open AI	invokeurl GET	API key	Supports AI-assisted responses to learners' queries.
Zoho Creator Data Integration	Zoho Creator forms and application data	Internal form operations / Deluge	Zoho Creator application access	Stores and manages course, topic, material, enrolment, subscription and tutoring information.

Code 5.3: Integration script

```
aiResponse = invokeurl
[
   url: "https://api.openai.com/v1/models"
   type: GET
   headers: {"Content-Type":"application/json"}
];

info aiResponse;
```

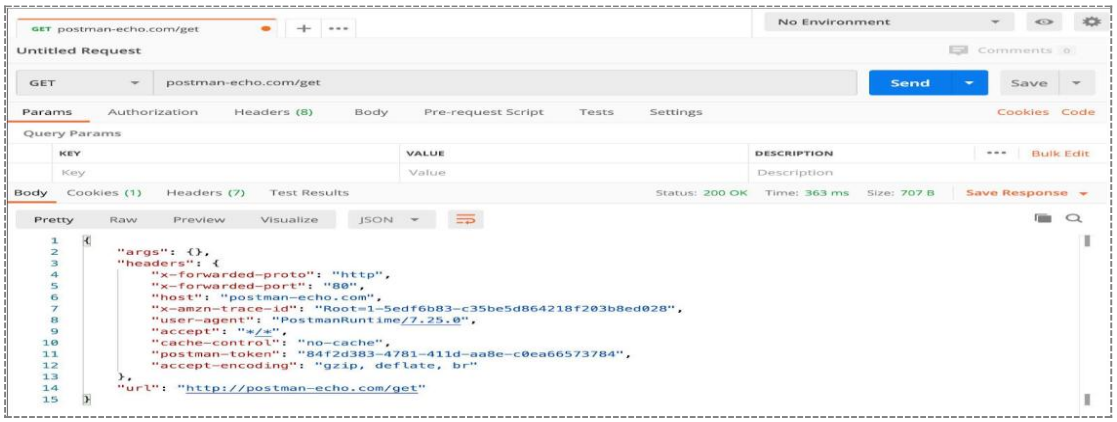


Figure 5.7: Connection Setup and Postman / Response Output

5.8 User Management and Role-Based Access

The SkillBridge application is designed to support different users through role-based access. The main user roles are Administrator, Student/Learner and Tutor/Manager.

- **Administrator:** Responsible for managing courses, categories, topics, learning materials, enrolment records, subscription and payment-related information. The administrator can also access the dashboard and available reports to monitor application activities.
- **Student/Learner:** Can access available courses and learning materials, submit enrolment details, view subscription-related information, raise tutoring requests and use the AI Learning Assistant.
- **Tutor:** Can review tutoring requests, process request-related activities and access tutoring approval and confirmation records.

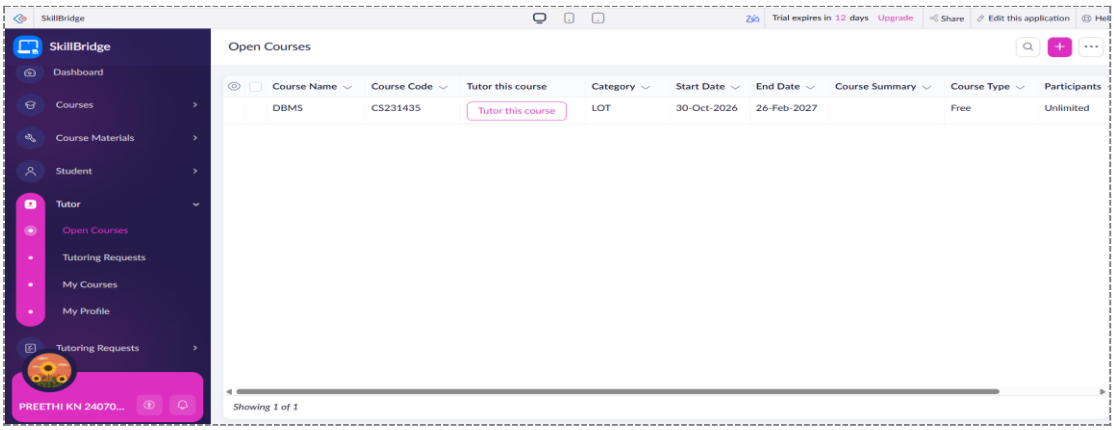


Figure 5.8: Tutor asking Admin to tutor the course

5.9 AI and Analytics Features

SkillBridge includes an AI Learning Assistant to support learners by providing AI-assisted responses to their learning-related queries. The AI functionality is connected through the configured Zoho CRM integration. Learners can submit questions and receive responses through the application.

The application also uses reports and dashboards to organize and present information about courses, course categories, topics, learning materials, student enrolments, subscriptions, payments and tutoring requests. These reports help administrators monitor learning activities, review records and manage the overall learning process.

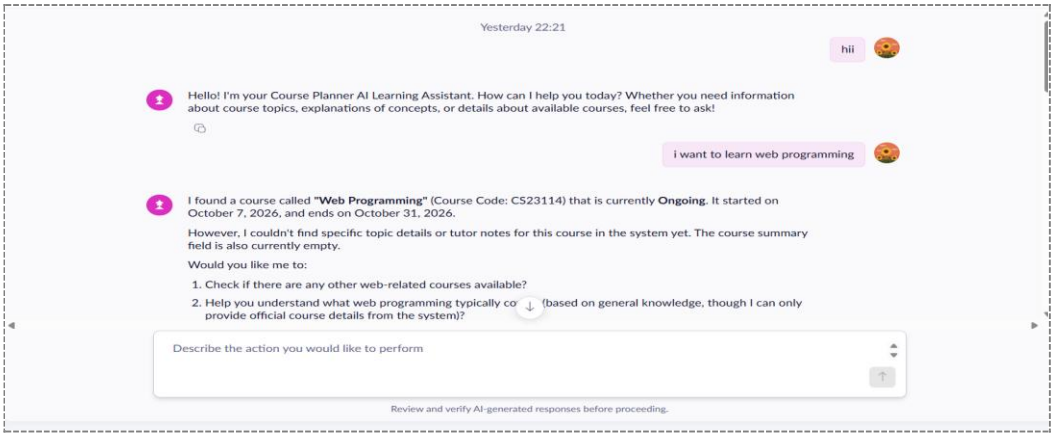


Figure 5.9: AI agent

5.10 Pro-Code Extension

SkillBridge was developed using Zoho Creator's low-code development environment, which provides built-in forms, reports, pages, workflows and Deluge scripting capabilities.

Advanced pro-code extensions such as Zoho Catalyst, JavaScript widgets and custom Creator REST API services were not separately implemented in the current version.

However, these technologies could be used in future enhancements. Zoho Catalyst could support additional backend services, JavaScript widgets could provide more interactive user interfaces, and Creator REST APIs could enable integration with external applications. These extensions would improve the application's flexibility, scalability and functionality

5.11 Mobile and Portal Access

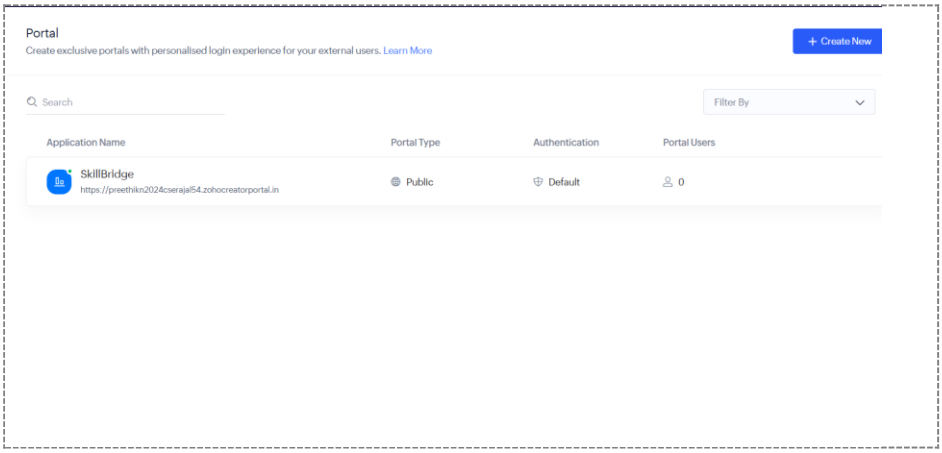


Figure 5.10: Customer Portal

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out as an important part of the SkillBridge application development process to verify that the forms, workflows, reports and integrations function as expected. The following testing approaches were considered:

1. Unit Testing

Individual forms and Deluge scripts were checked to verify that they accept valid inputs, validate mandatory fields and perform their intended operations. Forms such as Add Course, Add Topic, Add Course Material, Enrol User and Tutoring Request were included.

2. Integration Testing

The interaction between forms, workflows and the configured AI integration was considered. The AI Learning Assistant and its response-related workflow were checked to ensure that submitted queries could be processed through the configured integration.

3. Role-Based Access Testing

Application access was reviewed from the perspective of different users, including administrators, tutors/managers and students. The purpose was to ensure that users could access the application features intended for their roles.

4. User Acceptance Testing

The application was evaluated from a learner's and administrator's perspective to check whether the main operations, including course management, accessing learning materials, enrolment and tutoring requests, were understandable and convenient.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Status
TC-01	FR-01	Add a new course	Enter valid course details and submit	Course record should be created successfully	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Status
TC-02	FR-02	Validate mandatory course fields	Submit the course form without required details	Validation message should be displayed	Pass
TC-03	FR-03	Add a course category and topic	Enter valid category and topic details	Records should be saved successfully	Pass
TC-04	NFR-01	Test unauthorised access	Attempt to access a restricted feature using a user account without permission	Access should be restricted according to configured permissions	Pass

6.3 Debugging Deluge Scripts

Debugging was considered during the development of SkillBridge to identify and correct errors in Deluge scripts and workflow operations. The following techniques can be used:

1. **Info Statements:** The info statement is used to display variable values and execution results during script execution.
2. **Deluge Editor:** The editor's available execution and debugging facilities can be used to inspect script behaviour.
3. **Workflow Logs:** Workflow execution details can be reviewed to identify errors in form events and automated operations.
4. **Error Messages:** Form validation messages and execution errors help identify incorrect inputs and script issues.
5. **Integration Response Checking:** API responses can be inspected to identify problems with external service communication.

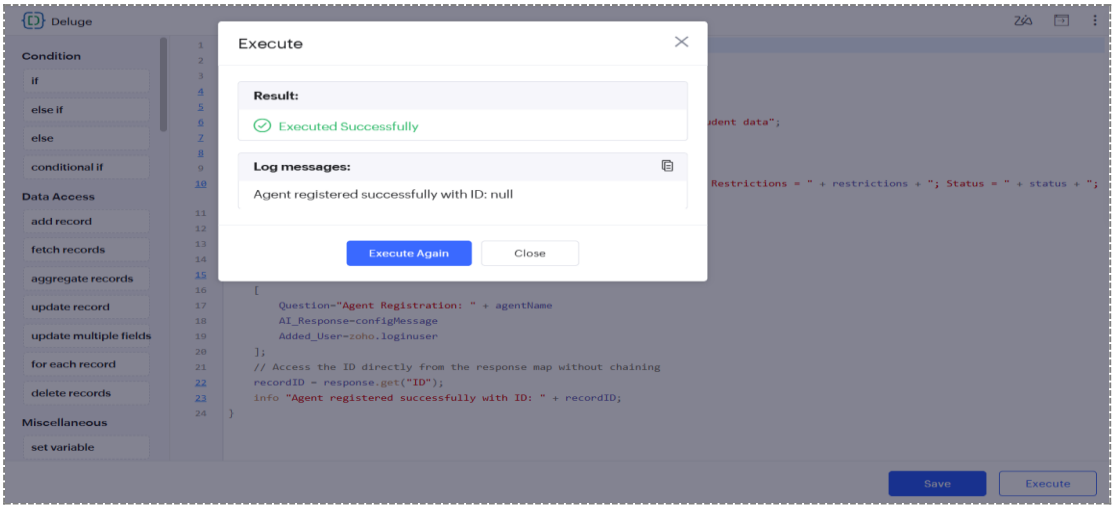


Figure 6.1: Deluge Debug Output / Log

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	AI Learning Assistant did not display the expected response	Response field or API response key was incorrectly mapped	Checked the API response and corrected the field mapping	Closed
BUG-02	Course form did not accept incomplete or incorrect input properly	Mandatory field validation was missing or incorrectly configured	Reviewed the form validation settings	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Student / Learner	Course details and learning materials were easy to access.	Reviewed the course and material navigation.
User 2	Administrator	Managing course information and viewing reports was convenient.	Reviewed the dashboard and report layout.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

SkillBridge was developed using the Zoho Creator platform. The application was configured using forms, reports, pages, workflows and Deluge functions within the Creator environment.

The deployment process involves preparing the application, checking the functionality of its forms and workflows, reviewing the configured integrations and publishing the application for authorised users

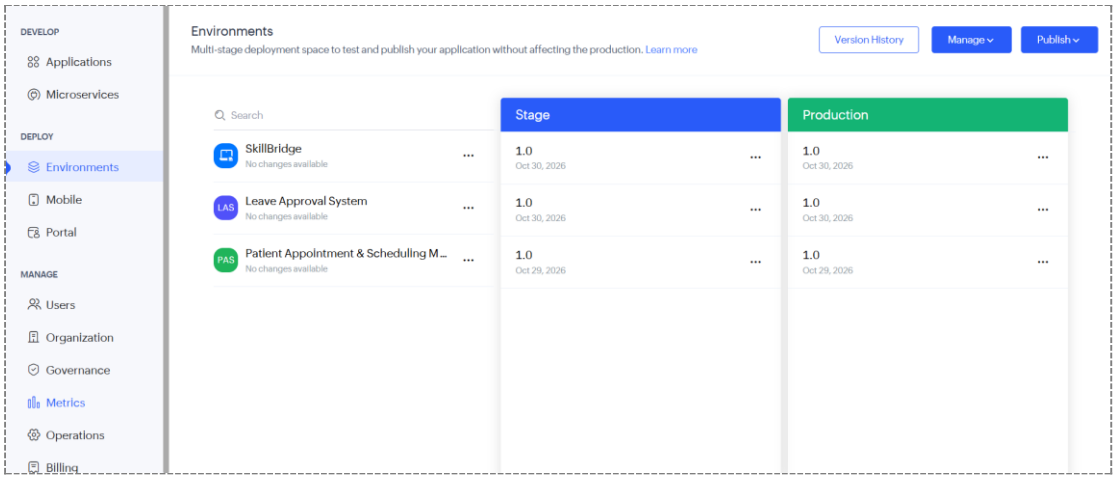


Figure 7.1: Environments / Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done
1	All test cases executed in Development / Stage	Done
2	Sample/test data removed or isolated	Done
3	Roles, permissions and users configured in Production	Done
4	Connections authorised in Production	Done
5	Schedules and workflows enabled	Done
6	Application published and URL verified	Done
7	Version / release notes recorded	Done

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Used report criteria to display relevant records	Course and Material Reports	All available records displayed	Relevant records displayed based on report criteria	Easier data access and reduced unnecessary record display
Organised learning materials into separate categories	Course Material Management	Materials accessed through general records	Materials organised by course and topic	Improved navigation and accessibility
Reused configured workflows for form operations	Form Workflows	Manual handling of repeated operations	Workflow-based processing	Reduced repetitive administrative work

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based access permissions	SkillBridge application access for administrators, tutors/managers and students	Unauthorised access to application features
Form and report access restrictions	Course, enrolment, subscription and tutoring forms and reports	Unauthorised viewing or modification of records
Field-level access restrictions	Sensitive fields, where configured	Exposure of confidential information

7.4 Governance and Administration

Governance and administration are important for maintaining the reliability, security and proper functioning of SkillBridge. The application is developed using Zoho Creator, which provides administrative features for managing application data, users and configurations.

The following practices are considered for SkillBridge:

1. Audit Trail: Application activities and changes can be monitored using the available Zoho Creator audit and activity tracking facilities. This helps administrators review important operations.

2. Change History: Changes made to forms, reports, pages and workflows should be reviewed and documented to maintain consistency and simplify troubleshooting.

3. Naming Conventions: Consistent names are used for application components. For example, forms use names such as Add_Course, Add_Topic and Enroll_User, making them easier to identify and maintain.

4. Backup and Export: Application data and configuration exports should be maintained periodically to support recovery and prevent data loss.

5. User and Licence Management: Application users and their access permissions should be reviewed regularly. Unnecessary access should be removed, and licences should be managed according to application requirements.

6. Application Maintenance: The administrator is responsible for reviewing application functionality, updating course information, checking workflows, monitoring integrations and resolving reported issues.

These governance practices help ensure that SkillBridge remains organised, secure and maintainable as its features and number of users grow.

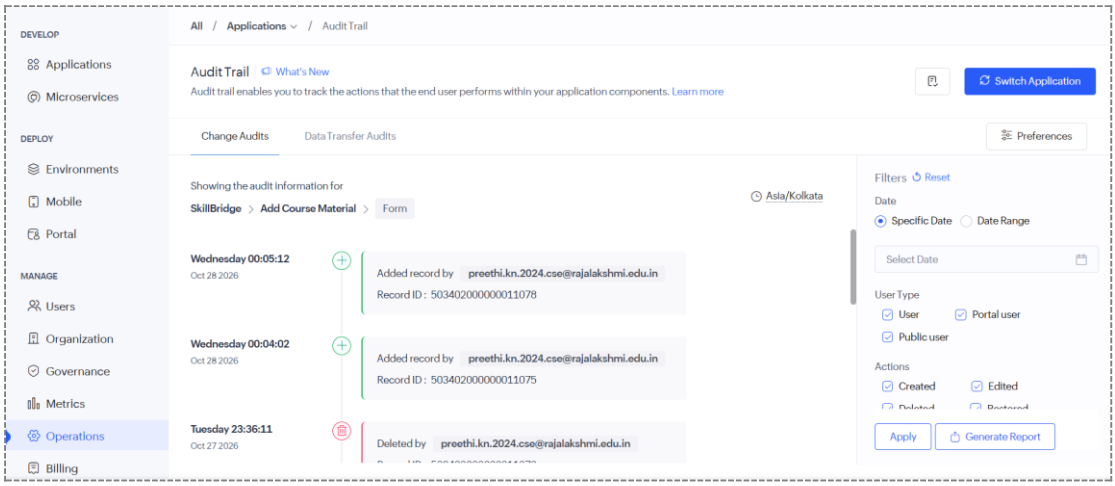


Figure 7.2: Audit Trail

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The SkillBridge application was developed to simplify skill development and competency management through a centralised learning platform. The following scenario demonstrates the major application operations from the administrator's and learner's perspectives.

Scenario: A student enrols in a course and accesses learning support.

Step 1: Course Management

The administrator adds course details, selects the appropriate course category and creates topics. This information forms the foundation of the learning platform.

Step 2: Course Enrolment and Learning Materials

The student enrols in an available course. The student can then access the associated learning materials through the application.

Step 3: AI Learning Assistance and Tutoring Support

The student submits a learning-related question through the AI Learning Assistant. The student can also submit a tutoring request when additional guidance is required.

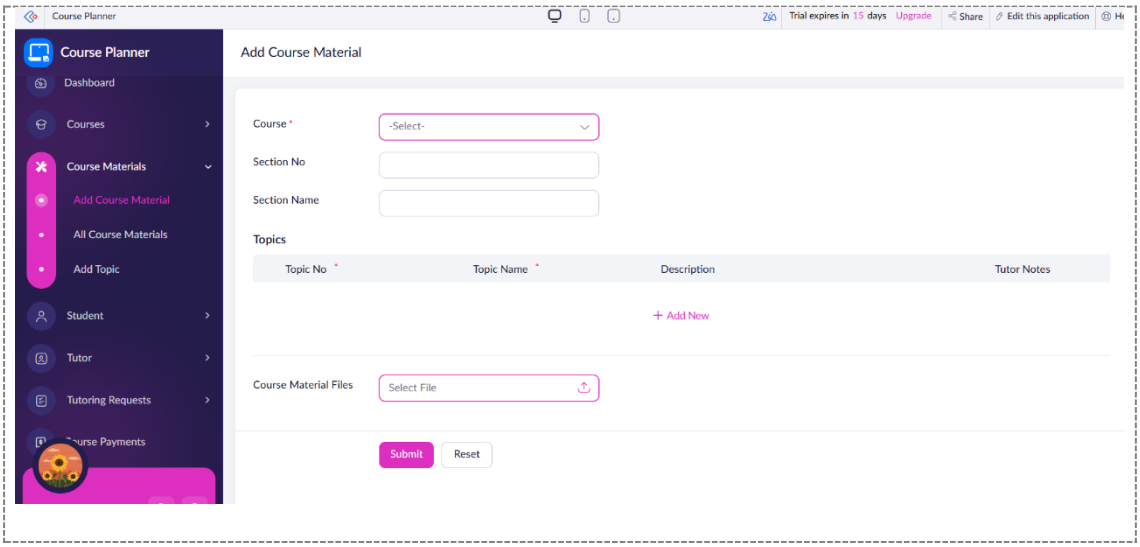


Figure 8.1: Scenario Step 1 – Course Management

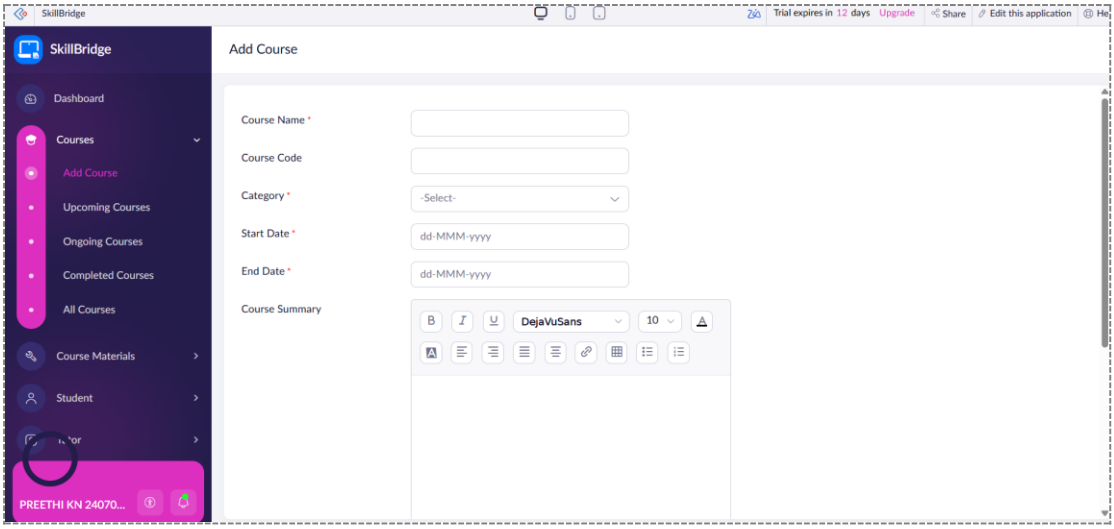


Figure 8.2: Scenario Step 2 – Course enrolment

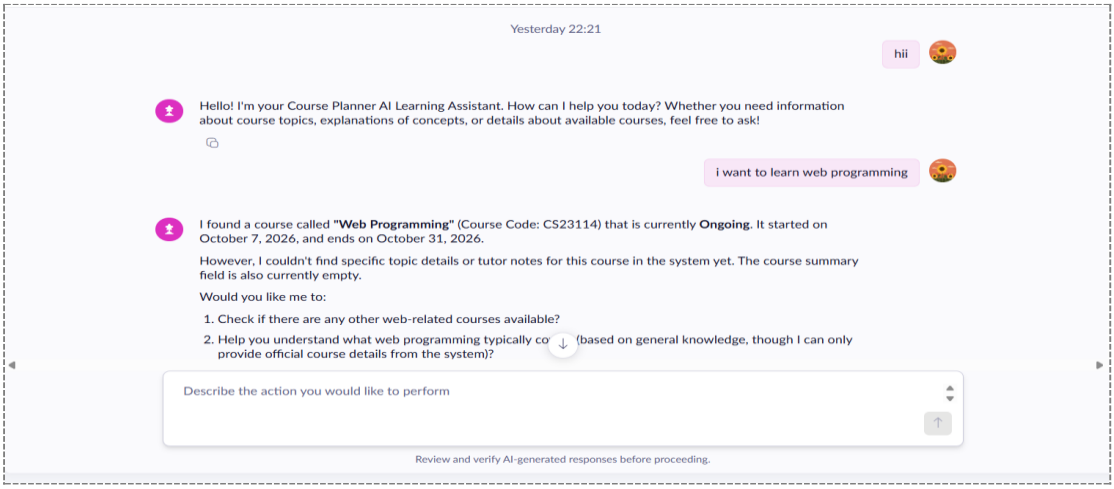


Figure 8.3: Scenario Step 3 – AI assistant

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
To develop a centralised platform for managing courses	Add Course, Add Course Category and Add Topic forms	Course information is organised within the application
To provide easy access to learning resources	Add Course Material form and All Course Materials page	Learning materials can be managed and accessed
To simplify student enrolment and subscription management	Enroll User, Course Subscription and Course Payment forms	Enrolment and subscription information can be recorded

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Course management	Course information may be maintained separately	Centralised course management	Better organisation of course information
Student enrolment	Enrolment details may be recorded manually	Enrolment records maintained through forms	Simplified record management
Learning materials	Resources may be shared through different platforms	Materials organised within SkillBridge	Easier access to learning resources
Report generation	Information may require manual compilation	Reports available within Zoho Creator	Easier monitoring of records

8.4 Limitations of the Low-Code Approach

Zoho Creator helped simplify the development of SkillBridge through its built-in forms, reports, pages and workflow features. However, some limitations of the low-code approach were identified.

1. **UI Customisation:** The platform provides predefined interface components, which can limit complete control over the design and layout. This can be addressed using custom pages and supported scripting options.
2. **Deluge Limitations:** Complex application logic may require additional scripting and careful handling of form workflows. Deluge functions were used for application-specific operations.
3. **API Limitations:** External integrations depend on available API limits and authentication configurations. Secured connections and efficient API usage can help manage these constraints.
4. **Complex Functionality:** Highly customised features may require technologies beyond the built-in low-code tools.

5. **Vendor Dependency:** SkillBridge depends on the Zoho Creator platform and its supported services. Moving the application to another platform may require redesigning forms, workflows and integrations.

Despite these limitations, Zoho Creator provided the necessary development features for implementing the current version of SkillBridge.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

SkillBridge demonstrates the use of different development approaches within the application development spectrum.

1. No-Code Development

Built-in Zoho Creator features were used to create forms, reports and application pages with minimal coding. These features helped establish the basic structure of SkillBridge.

2. Low-Code Development

Deluge scripting was used to support custom application logic, form workflows and AI-related functionality. This provided flexibility beyond the standard no-code features.

3. Pro-Code Development

Advanced technologies such as Zoho Catalyst, JavaScript widgets and custom REST API services were not separately implemented in the current version. They could be introduced in future versions to support more complex functionality and custom interfaces.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The SkillBridge application was developed using Zoho Creator to provide a centralised platform for skill development and competency management. It brings together course management, categories, topics, learning materials, student enrolment, subscription records, tutoring requests and an AI-assisted learning feature. By using forms, reports, pages, workflows and Deluge scripting, the project demonstrates how a low-code platform can simplify application development and organise learning-related activities in one system.

The project achieved its main objective of creating a structured learning management application while providing practical experience in low-code development, workflow automation and application integration. SkillBridge also demonstrates how Zoho Creator can be used to develop and extend business applications with minimal traditional coding.

9.2 Future Enhancements

- **Predictive analytics using Zia:** Introduce AI-powered analytics to identify learning patterns, recommend suitable courses and monitor learner progress.
- **Payment gateway integration:** Integrate a secure payment gateway to support real-time subscription payments, transaction verification and automated payment confirmations.
- **Catalyst-based microservices:** Use Zoho Catalyst to handle complex processing and backend operations through scalable services.
- **Personalised learning recommendations:** Recommend courses and learning materials based on learners' interests, skills and progress.
- **Progress tracking and assessments:** Add quizzes, assessments, certificates and progress dashboards to measure learning outcomes

9.3 Learning Outcomes and Career Relevance

Developing SkillBridge provided practical knowledge of low-code application development using Zoho Creator. I gained experience in designing forms, creating reports and pages, managing application data, building workflows and writing Deluge scripts. The project also improved my understanding of integrating external services and using automation to reduce manual work.

These skills are relevant to several career pathways discussed in Unit V. As a **citizen developer**, I can use low-code tools to create solutions for practical problems. As a **low-code developer**, I can design applications and extend their functionality through scripting. The experience is also useful for a **solutions consultant**, who analyses business requirements and converts them into technical solutions, and an **automation engineer**, who develops workflows to improve operational efficiency.

Overall, this project strengthened my problem-solving, application design and technical skills while introducing me to the growing opportunities in low-code development and digital transformation.

REFERENCES

[1] A. Bucaioni, A. Cicchetti, and F. Ciccozzi, "Modelling in low-code development: A multi-vocal systematic review," *Software and Systems Modeling*, 2022, doi: 10.1007/s10270-021-00964-0.

[2] U. Frank, P. Maier, and A. Bock, "Low code platforms: Promises, concepts and prospects: A comparative study of ten systems," University of Duisburg-Essen, 2021. doi: 10.17185/duepublico/75244.

[3] Zoho Corporation, "Zoho Creator Help Documentation." [Online]. Available: https://www.zoho.com/creator/help/. [Accessed: Oct. 30, 2026].

[4] Zoho Corporation, "Deluge Script Reference Guide." [Online]. Available: https://www.zoho.com/deluge/. [Accessed: Oct. 30, 2026].

[5] Zoho Corporation, "Zoho Creator Resources." [Online]. Available: https://www.zoho.com/creator/resources/. [Accessed: Oct. 30, 2026].

[6] Moodle, "Moodle Learning Management System." [Online]. Available: https://moodle.org/. [Accessed: Oct. 30, 2026].

[7] Coursera, "Online Courses and Credentials." [Online]. Available: https://www.coursera.org/. [Accessed: Oct. 30, 2026].

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	SkillBridge application	5.1	Application overview screenshot
II	Forms with ≥ 8 distinct field types	Y	Ten application forms	5.2	Form design screenshots
II	Lookup field (relationship between forms)	Y	Course, Category, Topic and Enrolment forms	4.2, 5.2	ER diagram
II	Subform (parent–child data)	Y	Check form configurations	4.2, 5.2	Form design screenshot
II	Field validations / layout rules	Y	Form workflows and field settings	5.2	Validation code
II	List report with filter, grouping, custom action	Y	Application reports (28 report definitions)	5.3	Report screenshots
II	Kanban / Calendar / Timeline / Map report	Y	Check report types	5.3	Report screenshot
II	Pivot table / chart report	Y	Check report types	5.3	Chart/report screenshot
II	Page / dashboard built with UI Builder	Y	Dashboard, All_Course_Materials_Page	5.4	Dashboard screenshot
II	Deluge – form events (validate / success / field actions)	Y	Form workflows	5.5	Deluge code screenshot

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Deluge – functions, fetch, loops, collections	Y	Add_AI_Agent_Microservice function	5.5	Code listing
II	Email / in-app notifications	Y	Check workflow actions	5.5, 5.6	Workflow screenshot
II	Form workflow	Y	Seventeen form workflow definitions	5.6	Workflow screenshots
II	Scheduled workflow	Y	Configured schedule definition	5.6	Schedule screenshot
II	Approval process	Y	Tutoring_Approval_Confirmation form	5.6	Approval configuration
II	Blueprint	N	Not established in the application details	5.6	-
II	Batch / report workflow	Y	Check report actions and workflows	5.6	Workflow screenshot
II	Testing & debugging evidence	Y	Test cases and debugging records	6	Test results
III	Integration with a Zoho service	Y	Zoho CRM integration (as reported)	5.7	Integration screenshot
III	Connection / REST API call (invokeurl)	Y	Check Deluge scripts and connections	5.7	API code screenshot
III	Webhook or Creator REST API	Y	Check integration settings	5.7	API configuration
III	Users, roles and permission profiles	Y	Check user and permission settings	5.8	Permission screenshot
III	Record-level access criteria / customer portal	Y	Check access and portal settings	5.8	Access configuration
III	Zia / AI feature	Y	AI_Learning_Assistant	5.9	AI assistant screenshot
III	Analytics (charts / Zoho Analytics)	Y	Check dashboard and analytics features	5.9	Analytics screenshot

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Performance optimisation applied & measured	N	Optimisation recommendations	7.2	Performance evidence
IV	Deployment across environments	Y	Deployment configuration	7.1	Deployment screenshot
IV	Security controls (encryption, OAuth, permissions)	N	Security and connection settings	7.3	Security screenshot
IV	Governance (audit trail, administration)	Y	Audit and administration settings	7.4	Audit trail screenshot
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not established as implemented	5.10	-
V	Low-code limitations & spectrum analysis	N	Low-code, no-code and pro-code discussion	8.4, 8.5	-
—	Mobile app / portal access	Y	Check mobile and portal configuration	5.11	Mobile/portal screenshot

Total features implemented: 28 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	**3 – High**	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	**3 – High**	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	**2 – Medium**	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	**2 – Medium**	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	**3 – High**	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Add AI Agent Microservice

```
// Script ID : DS-01
// Location  : Zoho Creator → Functions → Add_AI_Agent_Microservice
// Purpose   : Inserts an agent-configuration message into the AI Learning
Assistant form.

// Paste the complete actual function code here.
```

C.2 Generate AI response

```
// Script ID : DS-02
// Location  : AI_Learning_Assistant → Form Workflow →
Generate_AI_Response
// Purpose   : Handles the AI response-related process in the AI Learning Assistant.

// Paste the complete actual workflow code here.
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open Google Chrome or another web browser.
2. Enter the SkillBridge application URL.
3. Log in using your registered Zoho account credentials.
4. After successful login, the SkillBridge dashboard will appear.
5. Use the navigation menu to access the available forms, pages and reports.

D.2 Instructions for Administrator

- **Step 1:** Log in and open the SkillBridge dashboard.
- **Step 2:** Select **Add Course** to create and submit new course details.
- **Step 3:** Use **Add Course Category** and **Add Topic** to organise courses.
- **Step 4:** Select **Add Course Material** to add learning resources.
- **Step 5:** Review enrolment, subscription and payment-related records.
- **Step 6:** Open the available reports and Dashboard to monitor application information.
- **Step 7:** Review tutoring requests and their approval or confirmation records.

D.3 Instructions for Tutor

- **Step 1:** Log in using your registered credentials.
- **Step 2:** Browse the available courses.
- **Step 3:** Complete the enrolment process using the **Enroll User** form.
- **Step 4:** Open the **All Course Materials** page to access learning resources.
- **Step 5:** View available subscription and payment-related information.
- **Step 6:** Submit a tutoring request using the **Tutoring Request** form.
- **Step 7:** Open the **AI Learning Assistant** to submit learning-related queries.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Preethi KN	Register No.	240701398
Project Title	SkillBridge - Skill development & Competency Management	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Preethi KN	Register No.	240701398
Department / Section	CSE	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____